Lesson 3.7 Guided Notes

Session 1 · Customary vs. Metric Conversions

I can convert a measurement between the customary and metric systems using a given rate. · 6.RP.A.3d

Word / Spanish	Meaning — complete the blank	Example
conversion factor / factor de conversión	A fact that _____ two units.	1 m ≈ 3.28 ft
approximately / aproximadamente	About. The symbol ≈ means “about _____ to.”	1 kg ≈ 2.2 lb
reasonableness / razonabilidad	Checking that your answer’s _____ makes sense.	bigger unit, smaller number

Key idea

Use the fact the problem gives. Many facts are rounded, so the answer is an _____.

Changing to a smaller unit, _____. Changing to a bigger unit, _____.

Worked example

A bag weighs 44 pounds. About how many kilograms is that? (1 kg ≈ 2.2 lb)

The fact: 1 kg ≈ _____ lb.

A kilogram is heavier than a pound, so the number gets smaller: _____.

44 ÷ 2.2 = 20. Check: 20 × 2.2 = 44.

Answer: 44 lb ≈ _____ kg.

You try

1. Convert. (1 kg ≈ 2.2 lb, 1 in = 2.54 cm)
a. 3 kg ≈ _____ lb
b. 10 in = _____ cm

_____

2. A fun run is 5 km long. About how many miles is that? (1 mi ≈ 1.6 km)

_____

Sum it up: To check that my conversion is reasonable, I ask ___.

_____

Session 2 · Multi-Step System Conversions and Dosage

I can solve a multi-step problem that requires converting units before using a rate. · 6.RP.A.3d

Word / Spanish	Meaning — complete the blank	Example
multi-step / de varios pasos	Do two or more operations in _____.	convert, then multiply
dose / dosis	The _____ of medicine to give.	120 mg
per / por	For _____ one.	4 mg per kg

Key idea

First change the given amount to the unit the _____ uses.

Write the _____ after every step so you know which operation comes next.

Worked example

A cat weighs 11 lb. The dose is 3 mg per kg. The syrup has 5 mg in each mL. How many mL? (1 kg ≈ 2.2 lb)

Pounds to kilograms: 11 ÷ 2.2 = _____ kg.

Dose: 5 kg × 3 mg per kg = _____ mg.

Syrup: 15 mg ÷ 5 mg per mL = _____ mL.

Answer: The cat gets _____ mL of syrup.

You try

1. Complete each step. (1 kg ≈ 2.2 lb)
a. 44 lb ≈ _____ kg
b. 20 kg × 3 mg per kg = _____ mg

_____

2. A garden bed is 4 ft by 6 ft. One bag of soil covers 8 square feet. How many bags are needed?

_____

Sum it up: In a multi-step problem, I convert first because ___.

_____

Session 1 · Model, reason, and explain

Customary vs. Metric Conversions · 6.RP.A.3d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Use 1 kilogram ≈ 2.2 pounds. A package weighs 15.4 pounds. Find its approximate mass in kilograms. Explain the operation, approximation symbol, and a reverse check.

Kilograms	Approximate pounds
1	2.2
?	15.4

_____

_____

_____

Extend your reasoning

Why does exact division of the supplied numbers not make the real-world conversion exact?

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Multi-Step System Conversions and Dosage · 6.RP.A.3d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A craft mixture uses 3 grams of pigment per kilogram of clay. You have 11 pounds of clay. Use 1 kg ≈ 2.2 lb. Pigment is supplied at 5 grams per scoop. About how many scoops are needed? Show a path through the units.



_____

_____

_____

Extend your reasoning

Explain why rounding a conversion too early can change the final amount in another problem.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?